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PAPER

Non-invasive stimulation with temporal interference: optimization of the electric field deep in the brain with the use of a genetic algorithm

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E-mail: dstoupis@physics.auth.gr**Keywords:** temporal interference, deep brain stimulation, brain stimulation optimizationSupplementary material for this article is available [online](#)

Abstract

Objective. Since the introduction of transcranial temporal interference stimulation, there has been an ever-growing interest in this novel method, as it theoretically allows non-invasive stimulation of deep brain target regions. To date, attempts have been made to optimize the electrode montages and injected current to achieve personalized area targeting using two electrode pairs. Most of these methods use exhaustive search to find the best match, but faster and, at the same time, reliable solutions are required. In this study, the electrode combinations as well as the injected current for a two-electrode pair stimulation were optimized using a genetic algorithm, considering the right hippocampus as the region of interest (ROI). **Approach.** Simulations were performed on head models from the Population Head Model repository. First, each model was fitted with an electrode array based on the 10–10 international EEG electrode placement system. Following electrode placement, the models were meshed and solved for all single-pair electrode combinations, using an electrode on the left mastoid as a reference (ground). At the optimization stage, different electrode pairs and injection currents were tested using a genetic algorithm to obtain the optimal combination for each model, by setting three different maximum electric field thresholds (0.2, 0.5, and 0.8 V m^{-1}) in the ROI. The combinations below the set threshold were given a high penalty. **Main results.** Greater focality was achieved with our optimization, specifically in the ROI, with a significant decrease in the surrounding electric field intensity. In the non-optimized case, the mean brain volumes stimulated above 0.2 V m^{-1} were 99.9% in the ROI, and 76.4% in the rest of the gray matter. In contrast, the stimulated mean volumes were 91.4% and 29.6%, respectively, for the best optimization case with a threshold of 0.8 V m^{-1} . Additionally, the maximum electric field intensity inside the ROI was consistently higher than that outside of the ROI for all optimized cases. **Objective.** Given that the accomplishment of a globally optimal solution requires a brute-force approach, the use of a genetic algorithm can significantly decrease the optimization time, while achieving personalized deep brain stimulation. The results of this work can be used to facilitate further studies that are more clinically oriented; thus, enabling faster and at the same time accurate treatment planning for the stimulation sessions.

1. Introduction

The anatomical and functional complexity of the human brain makes the electrical stimulation of neurons located in specific areas a challenging task. Invasive stimulation is an obvious solution in some

cases; however, it poses considerable risk to the patient. Non-invasive methods have been proposed to stimulate specific areas deep in the brain to minimize the risk induced by invasive methods.

The most common non-invasive brain stimulation techniques include direct electric field induction

methods such as transcranial direct current stimulation (tDCS), transcranial alternating current stimulation, and magnetically induced electric field methods, namely, transcranial magnetic stimulation. The efficacy of such methods in specific psychiatric conditions ranges from promising to indisputable [1–4]. The main problem with all of the above is the depth at which stimulation is effective. Various targets, such as the thalamus or hippocampus, lie deep within the brain, and new ways to stimulate these areas non-invasively are yet to be fully developed. Such a method is the transcranial temporal interference stimulation (tTIS), initially proposed by Grossman *et al* [5]. This technique takes advantage of the spatial electromagnetic wave interference resulting from stimulation frequencies in the 1–100 kHz range. As neurons do not respond to higher than 1 kHz frequencies [6], the effect of the individual electrode high-frequency currents is minimal; thus, only the modulation envelope of the temporal interference stimulates the neurons (for a small frequency difference). The mechanisms by which this method works are still under active research, with the most recent developments being described by Mirzakhali *et al* [7] and Esmaeilpour *et al* [8].

There has been an increasing interest in the tTIS method, which, as Grossman *et al* [5] demonstrated, works well in murine subjects. More specifically, current research on this method seeks to optimize the electrode combinations and currents [9, 10]. Recognizing this problem in the present work, we aim to perform a study on the optimization effectiveness and present a way of optimizing tTIS stimulation of a specific target brain area in any head model that can be generated from suitable MRI data. The main deep brain areas investigated in this study were the hippocampus (H) and the thalamus (T). The hippocampus on the right hemisphere (*Hr*) was chosen as the target brain region of interest (ROI), since we can obtain results comparable to those of other studies [9, 10]. The thalamus, on the other hand, covers a significantly larger area than the hippocampus and lies in its close vicinity; therefore, it can be used as a control area for optimization, expecting to observe a significant suppression of the maximum electric field within the thalamic area, in the case of an acceptably optimized electric field in the unilateral hippocampal area.

Two different optimization assessment methods were used in this study. The first was the ratio of the electric field in a particular grey matter area to the electric field in the rest of the brain, called $R_{\text{area}}^{\text{Nh}}$. The second method was the comparison between the maximum modulated electric field in the ROI with that in the control region (right and left thalamus, *Tr* and *Tl* respectively) and the contralateral side of the ROI (the left hippocampus, *Hl*). The $R_{\text{area}}^{\text{Hr}}$ ratio is the optimization objective function, and it is presented in more detail in section 2.

2. Materials and methods

2.1. Simulation of tTIS electric field

For the tTIS stimulation technique, only the electric field is of interest. In the case of low-frequency current injected into the electrodes, the domain can be considered Ohmic quasi-static [11] without any significant displacement current [12]. This approximation simplifies the equations and, assuming that there are no internal current sources, the Laplace equation in the following form is solved in the finite element (FE) computational domain:

$$\nabla \cdot (\sigma \nabla \phi) = 0 \quad (1)$$

with σ being the electrical conductivity (S/m) of each tissue.

The conductivity value may vary depending on the frequency of the field or individual characteristics, such as age, gender, and pathology. The conductivity values proposed by McCann *et al* [13] were used in this study. Equation (1) was integrated over the entire volume, to calculate the electric potential at each node. All FE model (FEM) calculations were executed with SfePy [14] using the Portable, Extensible Toolkit for Scientific Computation (PETSc) package [15–17], with the conjugate gradient (CG) solver and an algebraic multigrid (BoomerAMG) matrix preconditioner from the High-Performance Preconditioners (HYPRE) [18]. Dirichlet boundary conditions were used on the electrodes, setting the active electrode to 1 V and the return electrode to −1 V. After the solution, the electric field values were scaled by the ratio of the desired electric current to the current passing through the electrode-skin interface.

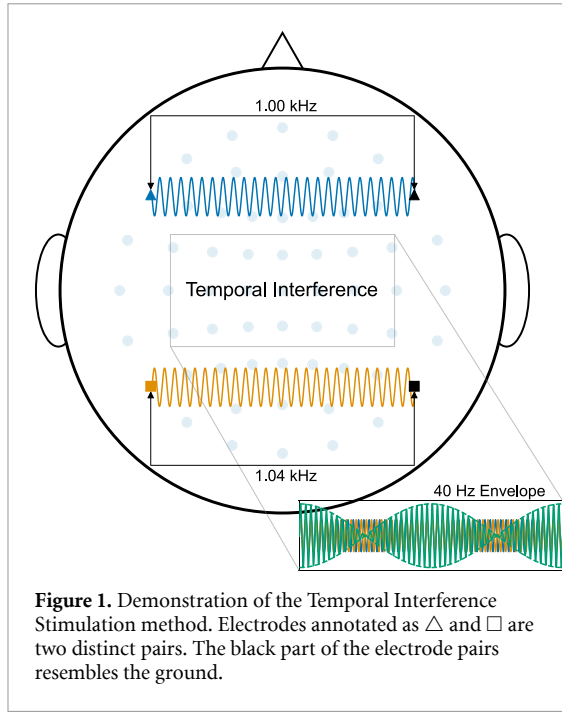
Using two pairs of electrodes (figure 1), with each pair having a slightly different frequency, an interference pattern can be generated in the conducting medium, whose envelope oscillates at the difference between the two frequencies. Depending on the nature of the medium, the pattern can vary, and as illustrated by Grossman *et al* [5], it can easily be calculated when a uniform medium is used.

Since the modulation occurs in 3D space, there will be different patterns in the x , y , and z directions. According to Grossman *et al* [5, p 20], the envelope amplitude of the amplitude modulation (AM) of the electric field produced by the temporal interference at any location $\vec{r} = (x, y, z)$, is calculated as

$$\vec{E}(\vec{n}, \vec{r}) = \left| |(\vec{E}_1 + \vec{E}_2) \cdot \vec{n}| - |(\vec{E}_1 - \vec{E}_2) \cdot \vec{n}| \right| \quad (2)$$

where $\vec{E}_1 = \vec{E}_1(\vec{r})$, $\vec{E}_2 = \vec{E}_2(\vec{r})$ are the electric fields coming from the two electrode pairs and $\vec{n} = \vec{n}(\vec{r})$ is the unit vector in the direction of interest.

The maximum amplitude of the modulation envelope at a specific location is of primary interest in this study because the modulation varies with time between zero and the maximum. Starting from



equation (2), the equations for calculating the maximum modulation amplitude across all directions at a specific location $\vec{r} = (x, y, z)$ are [5, p 20]:

$$E_{AM}^{\max}(\vec{r}) = \begin{cases} 2|\vec{E}_2| & \text{if } |\vec{E}_2| < |\vec{E}_1| \cos \alpha \\ 2 \frac{|\vec{E}_2 \times (\vec{E}_1 - \vec{E}_2)|}{|\vec{E}_1 - \vec{E}_2|} & \text{otherwise} \end{cases} \quad (3)$$

where α is the angle between $\vec{E}_1$ and $\vec{E}_2$, and equation (3) holds only if $\alpha < 90^\circ$.

When $\alpha \geq 90^\circ$, the sign of one of the two fields can be flipped (it must be done consistently) since reaching peak field strength at different time points across different areas violates the rule of less than 90° . This change is possible, considering that equation (3) calculates the maximum effect over one oscillation.

2.2. Modelling

2.2.1. Realistic head models

Realistic head models are required to generate accurate results for the field distribution; such models can be generated from MRI scans of individuals. Obtaining meaningful results requires the use of head models from several subjects. The Population Head Models (PHM) repository [3, 19], is a valuable resource with 50 head models from healthy individuals aged 22–35 years, with an anatomically realistic grey matter structure. Of these 50 models, 38 were used in this study, and their IDs are listed in table 1. The omitted models had insolvable self-intersections in their generated surface meshes, rendering numerical calculations impossible.

Each model comprised eight surface meshes: skin, skull, cerebrospinal fluid (CSF), gray matter, white matter, cerebellum, and ventricles. The tetrahedral

mesh of each model was generated using TetGen [20]. The meshing parameters for TetGen were chosen based on the study by Tran *et al* [21], taking into account the computational resources required. The final meshes were produced with $q = 1.4$ (tetrahedral radius-edge ratio) and $v_{\max} = 8$ (maximum individual tetrahedron volume).

2.2.2. Electrodes

All models were fitted with electrodes based on the international 10–10 EEG electrode placement system [22]. The four required fiducials (nasion, inion, and right and left preauricular), were individually identified for each model. Mesh2EEG [23, 24] was used to calculate the positions of the electrodes on the skin of each model, yielding 61 electrodes. An additional electrode was placed on the left mastoid, which served as a reference electrode (P9 in the international 10–10 system). In this study, only the electrodes shown in figure 2 were considered.

2.2.3. Automatic anatomical labeling (AAL) atlas registration

Atlas labeling is required to perform meaningful comparisons between different models because geometry varies significantly. To efficiently perform region analysis, the models were converted to the Neuroimaging Informatics Technology Initiative format and then normalized to the Montreal Neurological Institute (MNI) space using the Statistic Parametric Modelling library [25] in MATLAB (Mathworks, Natick, MA). Each model had a resolution of 1 mm per voxel, and after normalization, the individual brain areas were identified using the AAL atlas v3.1 [26].

2.3. Optimizing electrode montages and currents

Electrode position and current intensity optimization have been widely studied in tDCS [27, 28], although the problem in tTIS is not convex. Owing to the nonlinearity and non-convex nature of the tTIS method, exhaustive search methods [9, 10] are usually employed to achieve personalized optimization. We used an evolutionary algorithm to determine the optimal electrode and current combination for targeting (stimulating) a specific ROI in the brain.

2.3.1. Objective function

The fitness or objective function used to achieve optimization is given by the ratio shown in equation (4), which is calculated for each electrode combination and current combination:

$$R_{\text{area}}^{\text{Nh}} = \frac{\sum_i \frac{1}{V_i^{\text{Nh}}} E_i^{\text{Nh}}}{\sum_j \frac{1}{V_j^{\text{Rest}}} E_j^{\text{Rest}}} \quad (4)$$

where E_i^{Nh} , V_i^{Nh} and E_j^{Rest} , V_j^{Rest} are the electric field and volume in the selected grey matter area and the rest of the brain grey matter structures, respectively. Furthermore, the superscript of the ratio varies

Table 1. Index assignment to each model, based on the ID from the PHM repository. All the figures of this work will follow this convention, unless a different notation is explicitly used.

Index	Model ID	Index	Model ID	Index	Model ID
1	101309	14	124422	27	149337
2	103111	15	125525	28	151223
3	103414	16	126325	29	151627
4	105014	17	127630	30	153025
5	105115	18	128632	31	159340
6	110411	19	129028	32	160123
7	111716	20	130013	33	162733
8	113619	21	130316	34	163129
9	117122	22	131217	35	176542
10	118932	23	133019	36	178950
11	120111	24	133928	37	196750
12	122317	25	146432	38	198451
13	122620	26	147737		

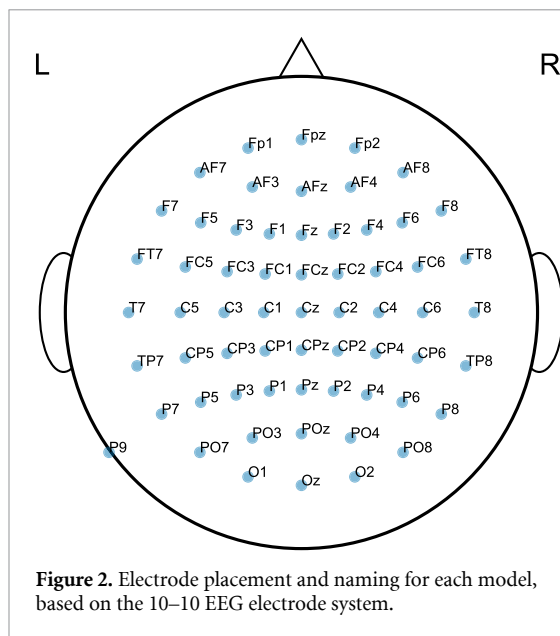


Figure 2. Electrode placement and naming for each model, based on the 10–10 EEG electrode system.

depending on the area considered. The naming convention for the superscript is to replace **N** with the first letter of the brain area and **h** with the hemisphere side, either right ($h = r$) or left ($h = l$), so in the case of the right hippocampus the ratio is denoted as $R_{\text{area}}^{\text{Hr}}$. (A short algorithm in the supplementary material shows the calculation of the fitness function for the optimization.)

The ratio described in equation (4) was chosen to maximize the electric field in the ROI and minimize it in the non-ROI regions (ideally zero). One helpful property of this ratio is that it is region volume independent, since the two volume-averaged electric fields take into account contributions from voxels all over the two volumes. Thus, the use of this ratio yields comparable values for all possible region combinations.

2.3.2. Solution search space and algorithm parameters
Before optimization, all models were numerically solved for the 61 electrodes using the extra electrode

on the left mastoid as the reference (figure 2). To minimize the search space and speed up the optimization process, the currents were chosen to have a sum of 2 mA, with a step of 0.05 mA, with the minimum and maximum current being 0.5 mA and 1.5 mA, respectively [10]. The genetic algorithm was initiated with random conditions (random pairs of electrodes) and had a population of 100 chromosomes (combinations), terminating at 25 generations (iterations). The elitist version of the algorithm was used, with the elitist ratio set to 0.01 [29].

Finally, a constraint was set on the maximum electric field in the target region (ROI). Each electrode combination stimulating below the set constraint threshold was assigned a large penalty [29] to steer the algorithm away from the low-intensity electric field values in the target. As discussed in [10], the set constraint values do not constitute a definite threshold. The effectiveness of the optimization method is presented in the Results and Discussion section for various values (0.2, 0.5, and 0.8 V m^{-1}) imposed as a maximum electric field constraint.

2.4. Data analysis

2.4.1. Non-optimized models

The non-optimized case refers to using the same electrode pairs and currents, regardless of the head model (human subject). The simulation results of this case were used to assess the performance of the optimization. The head models in the non-optimized case of this work were stimulated using the PO7-F7 and P8-FC6 electrode pairs (figure 2), and 1.25, 0.75 mA injected currents, respectively. These electrodes and current values were chosen based on the H_{sub2} optimized model of Lee *et al* [10].

2.4.2. Model correlation

The Pearson's correlation coefficient was calculated for each model, by correlating all models in a voxel-to-voxel scheme, for the non-optimized and optimized cases. The modulated electric field equation (3) was correlated for each voxel, after the models were

converted and fitted to the MNI space to eliminate the impact of the different geometry of the head models. With the correlation coefficient described here, we can assess the variability between single subjects, as described in Conta *et al* [30].

2.4.3. Focality and field distribution

In this study we used the modulated electric field value of 0.2 V m^{-1} as the stimulation threshold, based on Lee *et al* [10]. However, as Lee *et al* [10] discussed, the 0.2 V m^{-1} threshold is not definite as it potentially varies across individuals [31, 32]. This threshold is not to be confused with the optimization constraint of 0.2 V m^{-1} , as it refers to the threshold above which the electric field starts to have an effect on the neuron's behavior.

Establishing a focality measure can aid in assessing the quality of regional targeting in any of the cases discussed in this study. We introduce Vol_{80} , which indicates the total volume of the ROI where the condition $E_{\text{AM}}^{\text{max}} \geq 0.8 \cdot \max(E_{\text{AM}}^{\text{max}})$ is true.

Furthermore, we chose the nine best head models based on their $R_{\text{area}}^{\text{Hr}}$ ratio in descending order for the 0.8 V m^{-1} optimization threshold to illustrate the field distribution in the non-optimized and optimized cases. To assess the differences we introduce a new quantity named $d_{0.8}$ which is given by equation (5)

$$d_{0.8}^j = E_{\text{non-opt}}^j - E_{0.8}^j \quad (5)$$

where j is the voxel index, $E_{0.8}^j$ is the AM electric field for an optimization threshold of 0.8 V m^{-1} and $E_{\text{non-opt}}^j$ is the electric field in the non-optimized case. The threshold of valid values for the field distribution was set as $|d_{0.8}| \geq 0.2$, since the maximum electric field at or above 0.2 V m^{-1} is considered important in this work. The values that did not satisfy the threshold condition were set to zero.

2.4.4. Optimization effectiveness

To assess the optimization effectiveness, we generated box plots of the ratios $R_{\text{area}}^{\text{Hr}}$ (ROI), $R_{\text{area}}^{\text{Hl}}$, $R_{\text{area}}^{\text{Tr}}$, and $R_{\text{area}}^{\text{Th}}$, for the Hippocampus (H) and the Thalamus (T) in each hemisphere, for each model. Additionally, the maximum modulated electric field box plots were generated for the same areas. These plots aim to showcase the effect of the optimization in the right hippocampus (ROI) and in the areas of the immediate vicinity of the ROI, which is the thalamus.

2.4.5. Optimization variability

For the optimization variability assessment, line plots for all head models were generated. Each point represents one of the 38 models, with the horizontal axis indicating the model index and the vertical axis showing either the $R_{\text{area}}^{\text{Nh}}$ ratio or the maximum electric field, for the relevant plotted region. The vertical axis data were sorted in ascending order to make

the comparison easier. Therefore, for the same index in the horizontal axis, the values do not necessarily belong to the same model. (A 1-to-1 correlation of the points for the different optimization thresholds can be found in the supplementary material.)

3. Results and discussion

To comparatively assess the optimization results, the left and right regions of the thalamus and hippocampus were used. Since the ROI is the right hippocampus (Hr), stimulation in all other regions is expected to be minimal, that is, ideally zero. However, zero stimulation in the regions outside the ROI cannot be achieved with this method, as the resolution provided by the electrode montages is far less than that required to achieve pinpoint accuracy.

All model voxels (skin, skull, CSF, etc) other than those in the identified AAL areas were discarded, and data calculations were performed strictly using the AAL areas. The optimization results are presented with the 0.2 , 0.5 , and 0.8 V m^{-1} thresholds imposed for the genetic algorithm penalty function (figures 6–9).

3.1. Model correlation

It can be seen in figure 3 that for the non-optimized case (upper left subplot) there is some variability between single subjects, as described in Conta *et al* [30]. The variability in the optimized cases increases considerably, since the electrode montages and current settings are not the same for all subjects.

3.2. Focality and field distribution

Based on the 0.2 V m^{-1} stimulation threshold [10], a comparison can be made for all cases using the percentage of the brain volume stimulated at or above the 0.2 V m^{-1} maximum modulated electric field for the non-optimized and the 0.2 , 0.5 , and 0.8 V m^{-1} optimized cases. Figure 4 illustrates that there is a considerable reduction in the stimulated mean volume outside the ROI for all optimized cases, when compared to the non-optimized case. This reduction comes at a cost of a drop in the stimulated mean volume within the ROI. Although in the non-optimized case, on average 99.9% of the ROI is stimulated above the 0.2 V m^{-1} threshold, 76.4% of all other non-ROI areas are also stimulated. In contrast, there was a significant decrease in the stimulated volume outside the ROI in the optimized cases. The optimal stimulation, considering all cases, is achieved by the 0.8 V m^{-1} optimization threshold, which has the lowest spread (in the ROI) compared to all other thresholds, with the optimized case averaging a 91.4% ROI stimulation, while only 29.6% of non-ROI areas were stimulated.

Since the 0.8 V m^{-1} optimization threshold is the best achieved result, we will only consider this

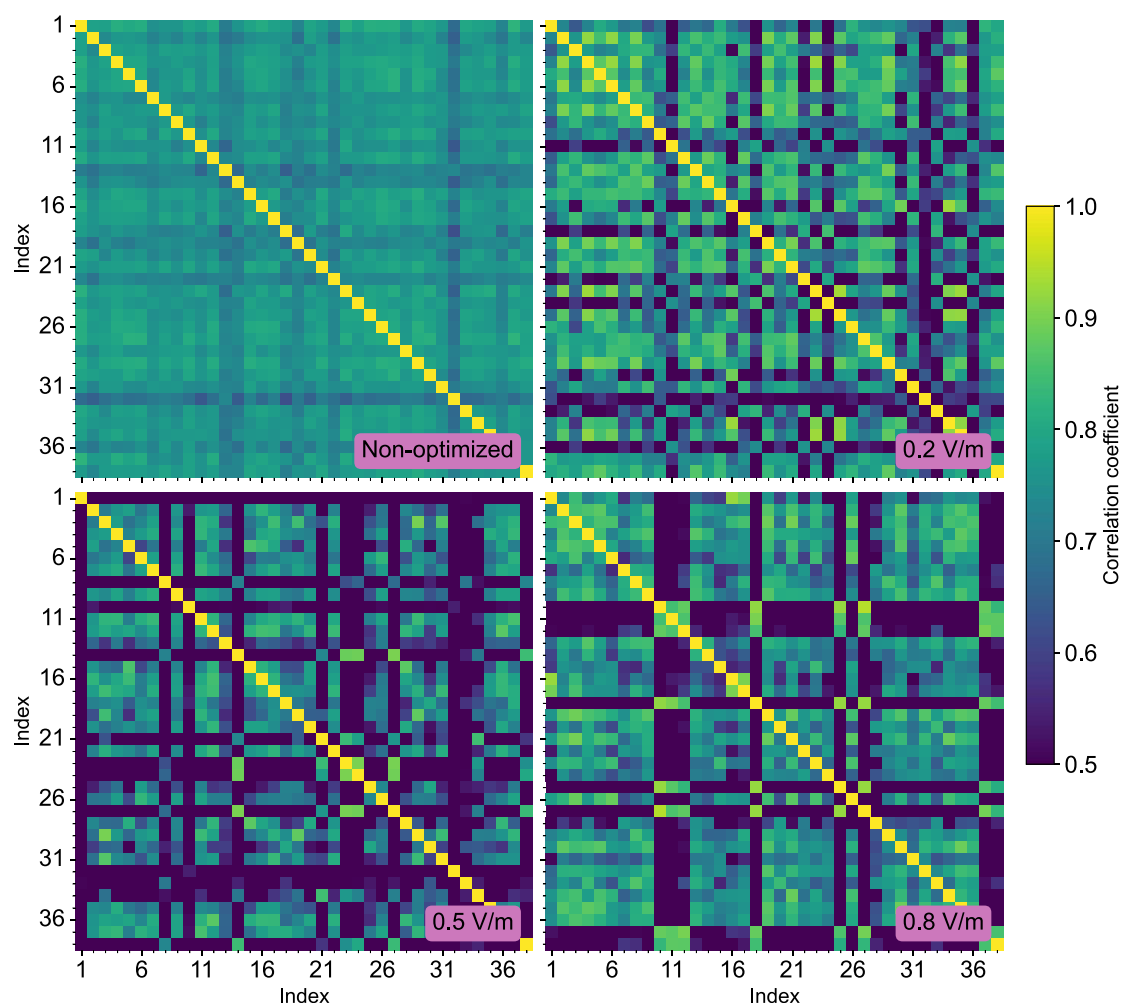


Figure 3. Correlation of the modulated electric field intensity across all models listed in table 1. Only the areas identified in the AAL atlas were considered in the correlation. The purple box at the lower right corner of each diagram indicates the optimization threshold. The colorbar shows the correlation coefficient.

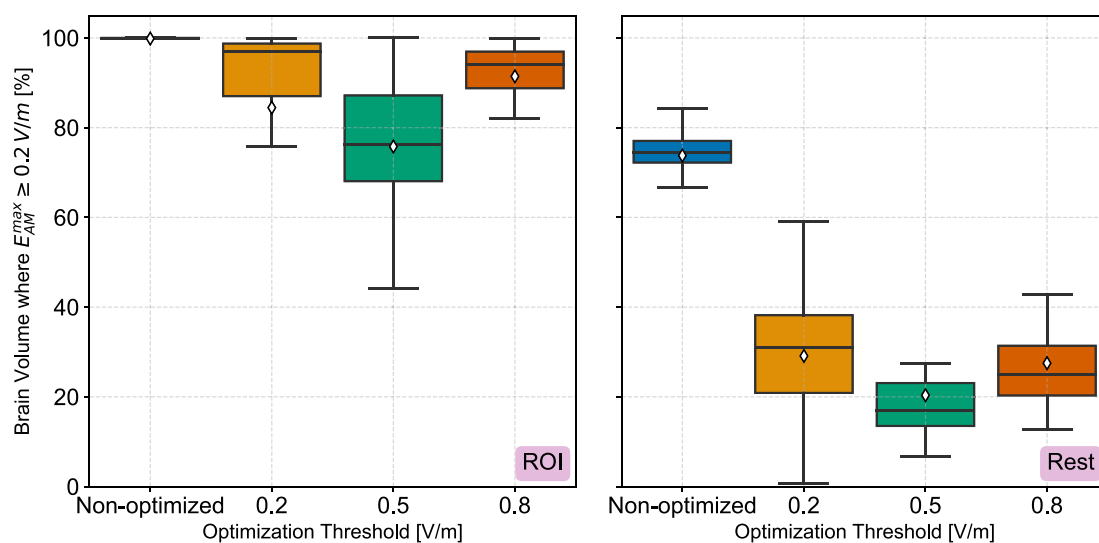
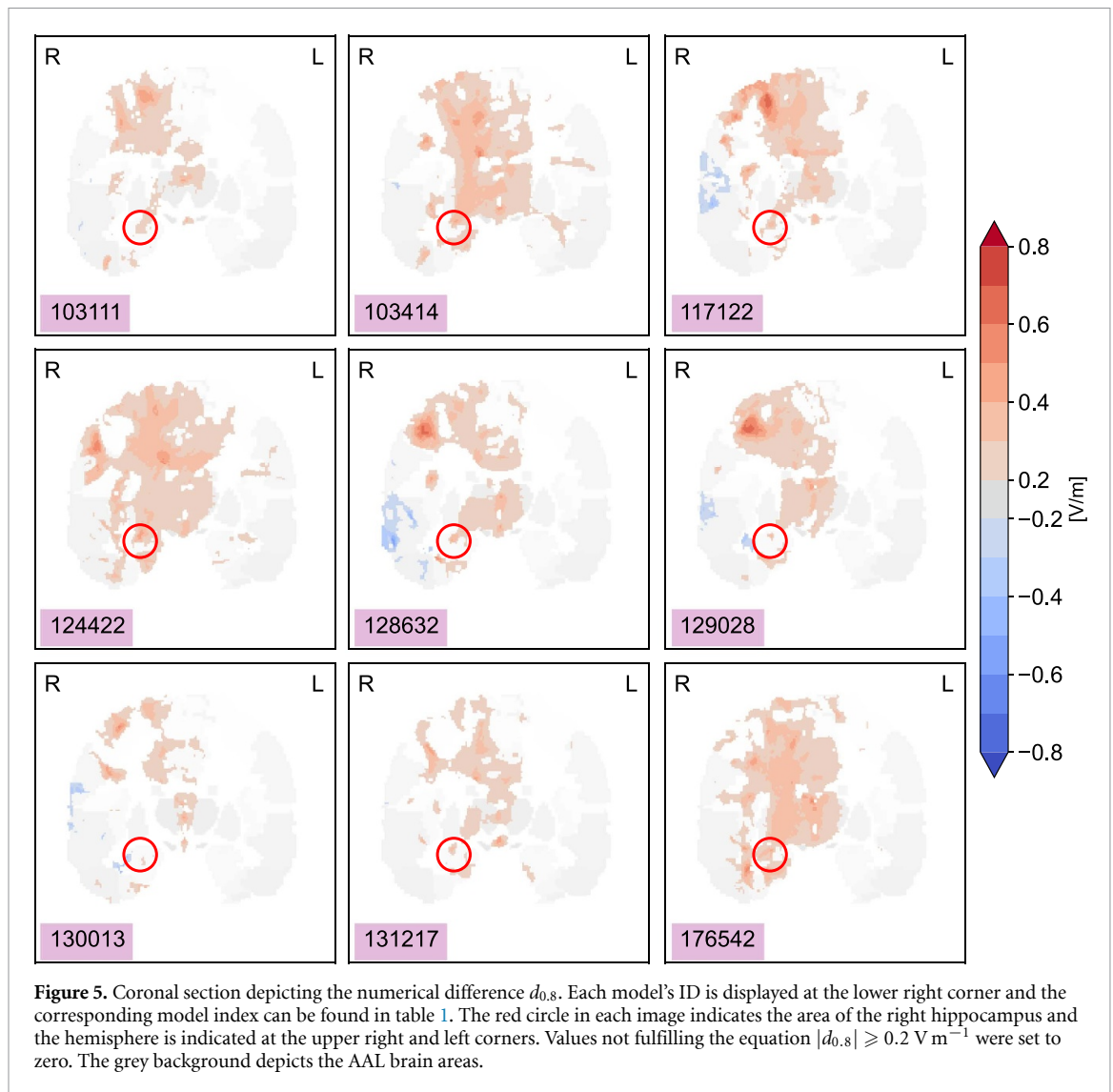


Figure 4. Percentage of the brain volume where $E_{AM}^{max} \geq 0.2 \text{ V m}^{-1}$ for the optimized and non-optimized stimulation of hippocampus, across all models. The subplot on the left shows the percentage inside the region of interest (hippocampus), while the subplot on the right shows the volume percentage on the rest of the regions, excluding the hippocampus. The $\diamond$ indicator represents the arithmetic mean.



threshold for Vol_{80} calculations. For all optimized models with the 0.8 V m^{-1} as the optimization threshold, the maximum Vol_{80} volume was 427 mm^3 (model index 24), the minimum was 35 mm^3 (model index 5), and the mean was 190.9 mm^3 . For comparison, in the non-optimized case, the maximum Vol_{80} volume was 443 mm^3 (model index 20), the minimum was 7 mm^3 (model index 12), and the mean volume was 74 mm^3 .

In figure 5, a coronal section of the numerical difference $d_{0.8}$ (equation (5)) is presented for the nine best head models. It is evident that even when optimization is applied, the variability in the pattern is not negligible. Furthermore, consistently for all nine models, the electric field in the non-optimized case has a greater spread in this particular slice, which is not the case for the optimized results. Thus, the focality of the field is significantly improved and figure 5 shows that the electric field is not sufficiently concentrated in the ROI. Considering that we used two electrode pairs in this study, focality resolution was limited. This problem can be addressed using more electrode pairs for stimulation [33].

3.3. Optimization effectiveness

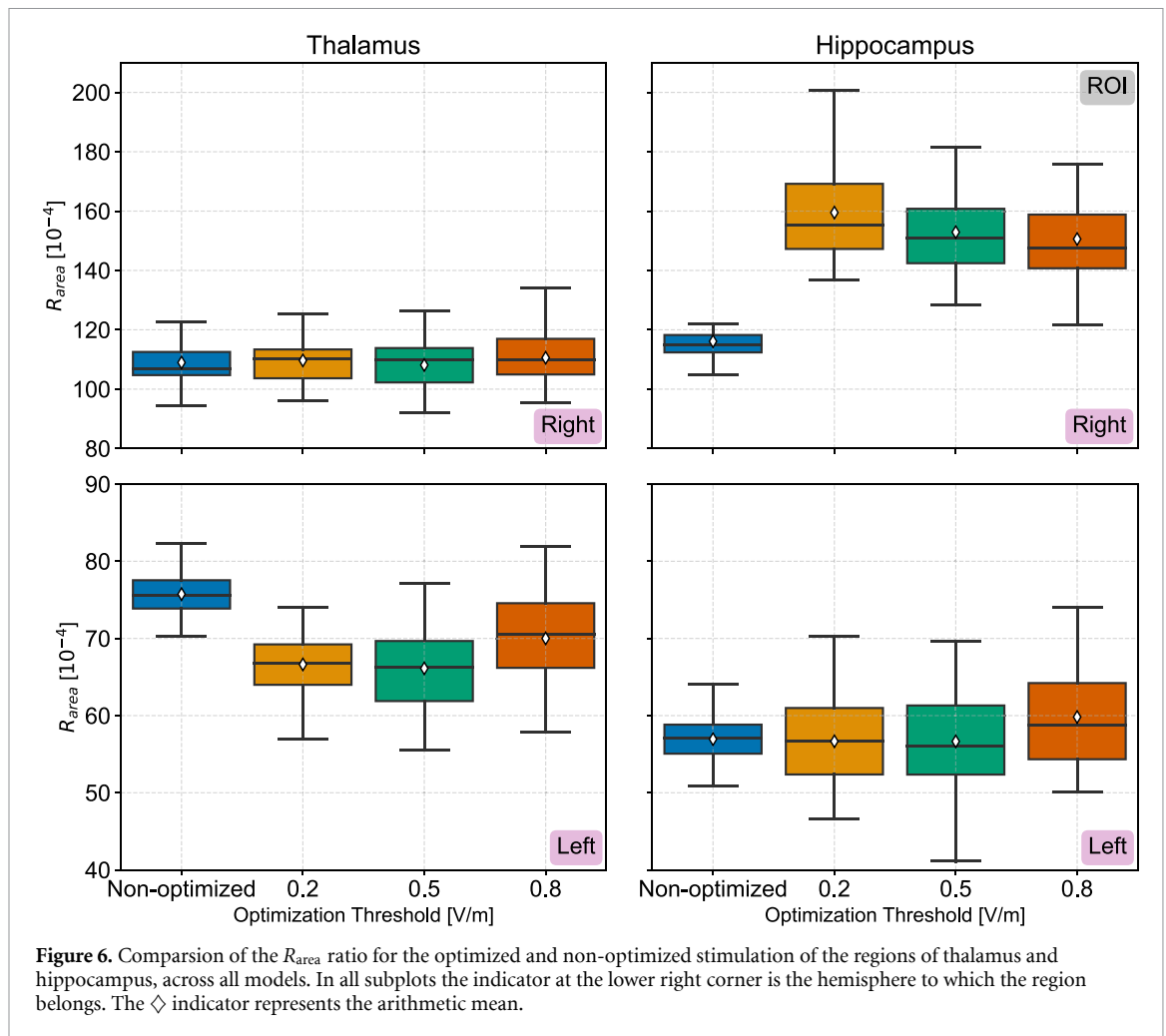
Figure 6 shows that $R_{\text{area}}^{\text{Nh}}$ does not vary considerably between different thresholds, indicating that the ability to focus on the electric field is independent of the maximum value achieved.

On the other hand, figure 7 shows that there is high variability in the maximum electric field when using the 0.2 V m^{-1} threshold. This behavior is expected since the modulated electric field deep in the brain is about 0.36 V m^{-1} per 1 mA of injected current [8]; thus, higher optimization thresholds will provide more stable and accurate optimization results.

3.4. Optimization variability

To better illustrate the optimization effect on each individual model used in this study, a comparison of the thalamus and hippocampus regions is presented in figures 8 and 9, respectively.

As shown in figure 8, using equation (4) as the optimization metric across all models, there is a minimal change in the different thresholds. In all optimized cases, the right hippocampus had a significantly



higher ratio than the left hippocampus, indicating a stronger field concentration in the right hippocampal area, which is also supported by figure 4.

Same as in figure 7, it can be observed in figure 9 that, as the threshold of the penalty function increases, the spread of the maximum value of the modulated electric field decreases. The wide variations in the right hippocampus for the 0.2 V m^{-1} threshold, can be attributed to the low threshold value, as the optimization algorithm is unable to successfully meet the optimization criteria. Since the total injected current from both electrode pairs is 2 mA , following the 0.36 V m^{-1} per 1 mA rule [8], the expected electric field values are larger than the constraint itself for the 0.2 and 0.5 V m^{-1} thresholds; thus, the algorithm may never approach an optimal solution.

Figure 9 indicates that stimulation in the region of the right hippocampus (Hr) can be achieved for all cases (non-optimized and optimized, alike), while the maximum electric field in the non-optimized case, for all other non-target regions, is almost double than the field of the optimized cases, which can lead to unwanted stimulation.

Optimization profoundly affects the maximum electric field value and distribution across the ROI.

The higher the optimization threshold, the smaller the spread of the maximum electric field observed between the models, leading to a more focused stimulation than the non-optimized stimulation.

3.5. Computational resources and time

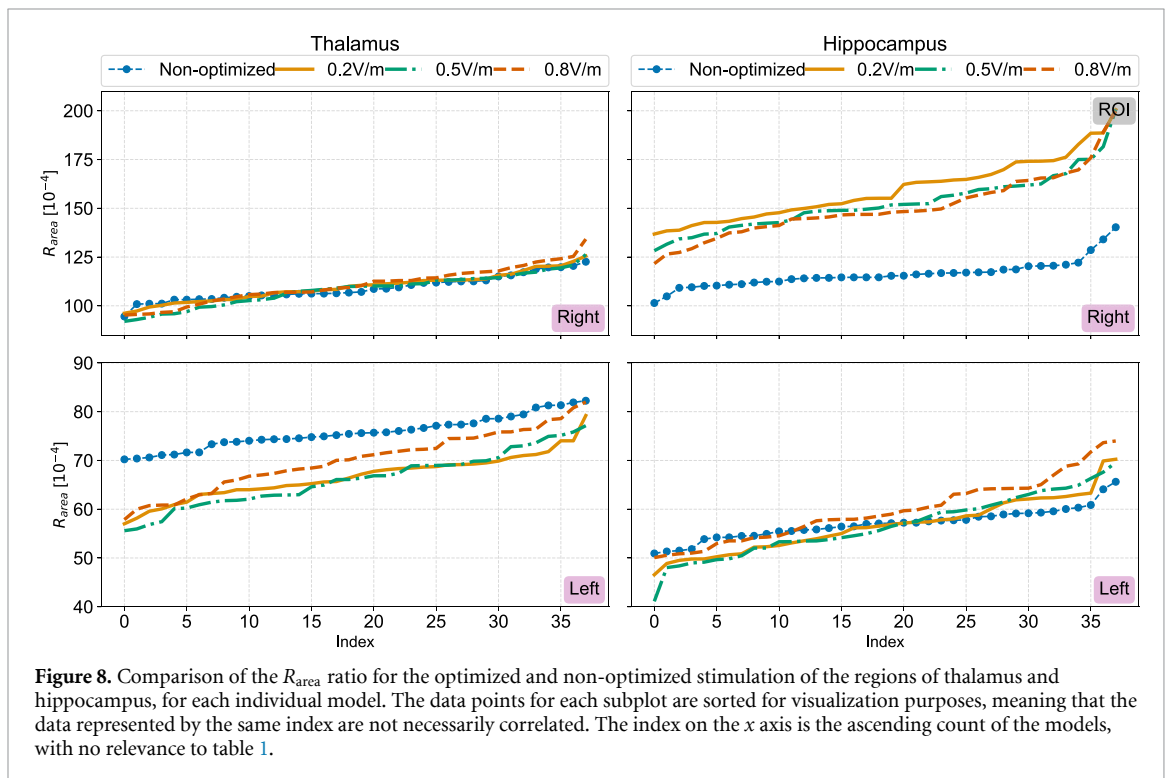
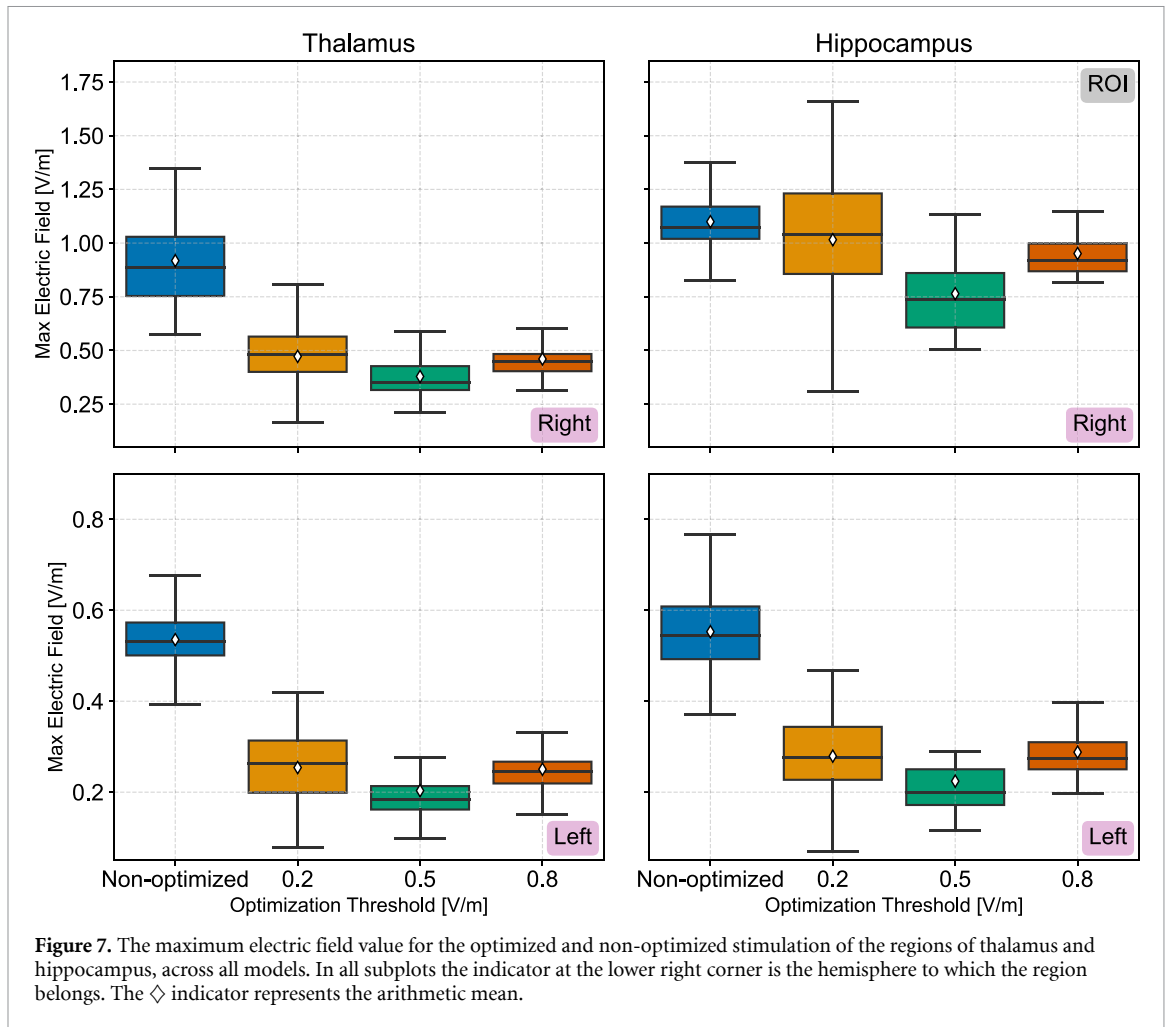
All simulations were performed on nodes with an Intel Xeon E5-2630 v4 processor (CPU) and 128GB of random access memory (RAM).

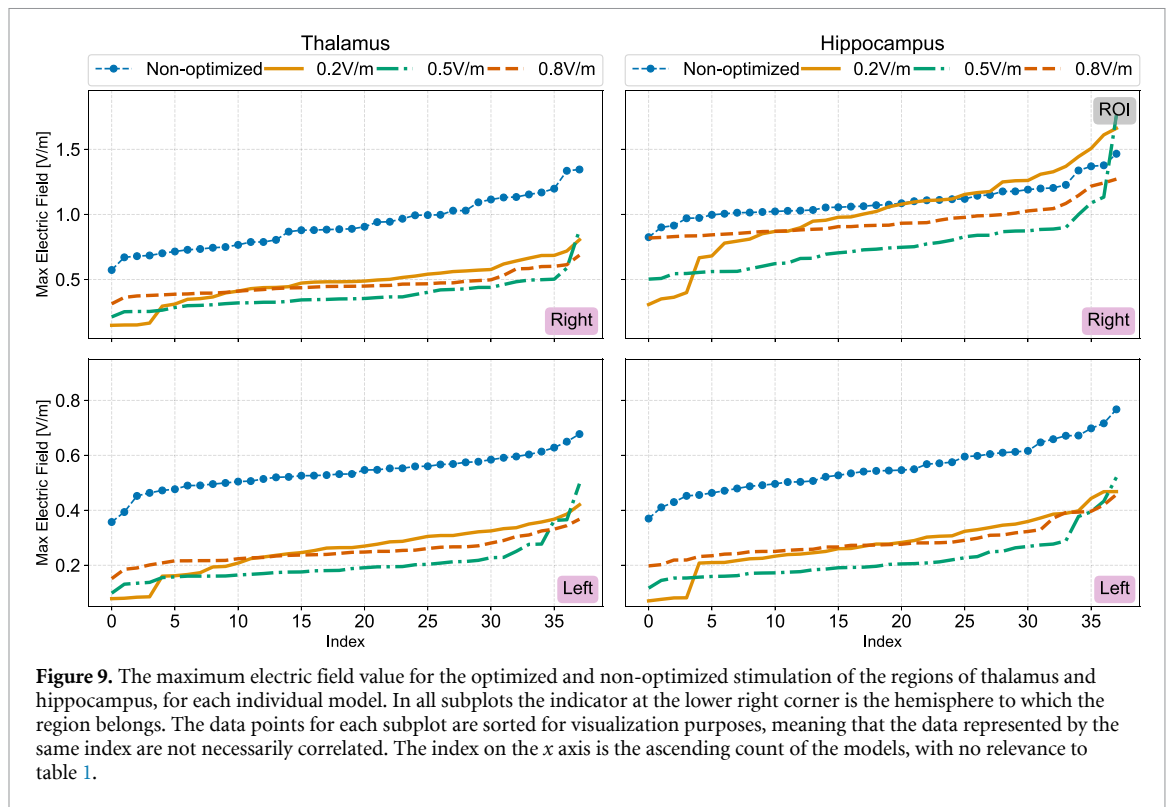
The total time required to perform an optimization for a single model was on average about 1.75 h by allocating 10 CPU cores and 32GB of RAM, when using the settings described in the subsection ‘Solution Search Space and Algorithm Parameters’.

Given that the accomplishment of a globally optimal solution consistently requires a brute-force approach, which is presently the best practice and can take up to 10 h [9, 10], employing the genetic algorithm proposed here can significantly decrease the optimization time and converge in less than 2 h for each model; thus accelerating the procedure by a factor of 5.

3.6. Method improvements and considerations

There have been studies regarding the robustness and effectiveness of the tTIS method in trying to resolve





focality issues. A multi-channel tTIS study on mice [33] indicated that targeting accuracy significantly improves when considering a six-channel (six electrode pairs) tTIS, performing 70.2% better than the traditional (two electrode pairs) tTIS. Furthermore, in [33], stimulation targeting appeared to be less affected when selecting different electrode pairs. In summary, Song *et al* [33] showed that more channels make the tTIS method more independent of the exact electrode placement while increasing focality. Studying the multi-channel approach on human head models and ultimately conducting clinical trials will shed light on potential improvements.

Additionally, as discussed and illustrated in [9], there is strong evidence [34, 35] that neurons respond preferentially to stimulation and that preference seems to be oriented along the neuron's axon direction. Comparing the field distribution in figure 5 with figures 5 and 11 from [9], it is evident that when the neuron stimulation preference is not considered, the field has a more extensive spread within the brain. Combining the neurons' preferential nature and the multi-channel approach, there are strong indications that the results will be significantly better at the individual level.

Moreover, as it is evident from equation (4), there are no margins set for the percentage of the ROI area that is considered important, nor are there margins set for the non-ROI areas. We wanted to test the ideal case where 100% of the ROI area is considered for optimization to assess the behavior of the tTIS in the best case, which is not necessarily a clinically correct approach. In future studies, such margins should be

considered, as they will yield more realistic results regarding specific ROI areas, such as the head of the hippocampus.

4. Conclusion

In this study, we applied the temporal interference stimulation (tTIS) method, using two electrode pairs, on 38 numerical head models of the PHM repository. Each model was fitted with electrodes according to the 10–10 international EEG electrode placement system. We optimized the electrode combinations and the injected current on each pair and showed that it is possible to achieve targeted stimulation on individual models. More specifically, from our analysis it is evident that greater focality was achieved, considering the right hippocampus as the ROI, with a significant decrease in the stimulated mean volume outside the ROI of about 46%, at a cost of a 9% drop within the ROI. At this point, it should be noted that our optimization approach is not of the kind 'one-size-fits-all': There might be clinical situations that require the whole ROI volume to be stimulated, irrespective of the stimulation that could take place in other brain areas. In such a situation, the above drop in the stimulated ROI volume could be considered important. However, in this work we have chosen an optimization approach that maximizes the stimulation volume in the ROI and suppresses the stimulation volume of other brain areas to avoid unwanted effects of the technique.

In addition, a diminishing spread of the maximum modulated electric field values in the

optimized models was observed as the optimization threshold increased. In contrast, in the non-optimized case (where the same electrode pairs and currents are used for all models), the modulated electric field across the models has relatively low variability, and region targeting is not optimal. Additionally, we observed that the selected objective function value for the optimization runs appeared to be independent of the maximum electric field value achieved in the target region. In some cases, the electric field outside the ROI is comparable to that inside the ROI, suggesting that achieving focused stimulation in all subjects with the tTIS method is difficult, mainly because of the small target volume.

Apart from purely anatomical constraints, the use of two electrode pairs is also a limiting factor because the electric field outside the target region is not easily steerable. Potentially utilizing more electrode pairs can solve the focality issues. Finally, despite the limitations, it seems that our approach reduces the optimization time significantly when compared to the current best-practice of exhaustive search, while allowing for targeted stimulation for each model.

Software availability

The software used in this publication are available on Zenodo [36].

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Conflict of interest

The authors have no conflicts of interest to declare.

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References

- [1] Jog M V, Wang D J and Narr K L 2019 A review of transcranial direct current stimulation (tDCS) for the individualized treatment of depressive symptoms *Personalized Med. Psychiatry* **17–18** 17–22
- [2] Brunoni A R *et al* 2016 Transcranial direct current stimulation for acute major depressive episodes: meta-analysis of individual patient data *Br. J. Psychiatry* **208** 522–31
- [3] Lee E G, Duffy W, Hadimani R L, Waris M, Siddiqui W, Islam F, Rajamani M, Nathan R and Jiles D C 2016 Investigational effect of brain-scalp distance on the efficacy of transcranial magnetic stimulation treatment in depression *IEEE Trans. Magn.* **52** 1–4
- [4] Moffa A H *et al* 2020 Efficacy and acceptability of transcranial direct current stimulation (tDCS) for major depressive disorder: an individual patient data meta-analysis *Progress Neuro-Psychopharmacol. Biol. Psychiatry* **99** 109836
- [5] Grossman N *et al* 2017 Noninvasive deep brain stimulation via temporally interfering electric fields *Cell* **169** 1029–41.e16
- [6] Hutcheon B and Resonance Y Y 2000 oscillation and the intrinsic frequency preferences of neurons *Trends Neurosci.* **23** 216–22
- [7] Mirzakhali E, Barra B, Capogrosso M and Lempka S F 2020 Biophysics of temporal interference stimulation *Cell Syst.* **11** 557–72.e5
- [8] Esmailpour Z, Kronberg G, Reato D, Parra L C and Bikson M 2021 Temporal interference stimulation targets deep brain regions by modulating neural oscillations *Brain Stimul.* **14** 55–65
- [9] Rampersad S, Roig-Solvas B, Yarossi M, Kulkarni P P, Santarnecchi E, Dorval A D and Brooks D H 2019 Prospects for transcranial temporal interference stimulation in humans: a computational study *NeuroImage* **202** 116124
- [10] Lee S, Lee C, Park J and Im C H 2020 Individually customized transcranial temporal interference stimulation for focused modulation of deep brain structures: a simulation study with different head models *Sci. Rep.* **10** 11730
- [11] Plonsey R and Heppner D B 1967 Considerations of quasi-stationarity in electrophysiological systems *Bull. Math. Biophys.* **29** 657–64
- [12] Barchanski A, Gersem H D, Gjonaj E and Weiland T 2005 Impact of the displacement current on low-frequency electromagnetic fields computed using high-resolution anatomy models *Phys. Med. Biol.* **50** N243–9
- [13] McCann H, Pisano G and Beltrachini L 2019 Variation in reported human head tissue electrical conductivity values *Brain Topogr.* **32** 825–58
- [14] Cimrman R, Lukeš V and Rohan E 2019 Multiscale finite element calculations in python using SfePy *Adv. Comput. Math.* **45** 1897–921
- [15] Balay S *et al* 2019 PETSc users manual *Technical Report ANL-95/11—Revision 3.11* (Argonne National Laboratory)
- [16] Balay S *et al* 2019 PETSc web page (available at: www.mcs.anl.gov/petsc)
- [17] Balay S, Gropp W D, McInnes L C and Smith B F 1997 Efficient management of parallelism in object oriented numerical software libraries *Modern Software Tools in Scientific Computing* ed E Arge, A M Bruaset and H P Langtangen (Boston, MA: Birkhäuser Press) pp 163–202
- [18] Lawrence Livermore National Laboratory HYPRE: high performance preconditioners (available at: <https://computation.llnl.gov/projects/hypre-scalable-linear-solvers-multigrid-methods>)
- [19] Erik G and Lee R L H 2016 Population head model repository (<https://doi.org/10.13099/VIP-PHM-V1.0>)
- [20] Hang S 2015 Tetgen, a Delaunay-based quality tetrahedral mesh generator *ACM Trans. Math. Softw.* **41**
- [21] Tran A P, Yan S and Fang Q 2020 Improving model-based functional near-infrared spectroscopy analysis using mesh-based anatomical and light-transport models *Neurophotonics* **7** 1
- [22] Nuwer M R 2018 10–10 electrode system for EEG recording vol 129
- [23] Paolo G, Katherine L P and Solomon G Diamond Mesh2EEG (available at: <https://engineering.dartmouth.edu/multimodal/mesh2eeg.html>)
- [24] Giacometti P, Perdue K L and Diamond S G 2014 Algorithm to find high density EEG scalp coordinates and analysis of

- their correspondence to structural and functional regions of the brain *J. Neurosci. Methods* **229** 84–96
- [25] Penny W, Ashburner J, Kiebel S, Henson R, Glaser D, Phillips C and Friston K S P M (available at: www.fil.ion.ucl.ac.uk/spm/)
- [26] Rolls E T, Huang C C, Lin C P, Feng J and Joliot M 2020 Automated anatomical labelling atlas 3 *NeuroImage* **206** 116189
- [27] Dmochowski J P, Datta A, Bikson M, Su Y and Parra L C 2011 Optimized multi-electrode stimulation increases focality and intensity at target *J. Neural Eng.* **8** 046011
- [28] Khorrampanah M, Seyedarabi H, Daneshvar S and Farhoudi M 2020 Optimization of montages and electric currents in tDCS *Comput. Biol. Med.* **125** 103998
- [29] Bozorg-Haddad O, Solgi M and Loáiciga H A 2017 *Meta-Heuristic and Evolutionary Algorithms for Engineering Optimization* (New York: Wiley)
- [30] von Conta J, Kasten F H, Ćurčić-Blake B, Aleman A, Thielscher A and Herrmann C S 2021 Interindividual variability of electric fields during transcranial temporal interference stimulation (tTIS) *Sci. Rep.* **11** 20357
- [31] Krause M R, Vieira P G, Csorba B A, Pilly P K and Pack C C 2019 Transcranial alternating current stimulation entrains single-neuron activity in the primate brain *Proc. Natl Acad. Sci.* **116** 5747–55
- [32] Rampersad S M, Janssen A M, Lucka F, Aydin U, Lanfer B, Lew S, Wolters C H, Stegeman D F and Oostendorp T F 2014 Simulating transcranial direct current stimulation with a detailed anisotropic human head model *IEEE Trans. Neural Syst. Rehabil. Eng.* **22** 441–52
- [33] Song X, Zhao X, Li X, Liu S and Ming D 2021 Multi-channel transcranial temporally interfering stimulation (tTIS): application to living mice brain *J. Neural Eng.* **18** 036003
- [34] Rushton W A H 1927 The effect upon the threshold for nervous excitation of the length of nerve exposed and the angle between current and nerve *J. Physiol.* **63** 357–77
- [35] Radman T, Datta A, Ramos R L, Brumberg J C and Bikson M 2009 One-dimensional representation of a neuron in a uniform electric field *2009 Annual Int. Conf. IEEE Engineering in Medicine and Biology Society* pp 6481–4
- [36] Stoupis D 2022 dimst23/tTIS-FEM-and-Optimization: tTIS FEM and optimization code (<https://doi.org/10.5281/zenodo.5907212>)